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#include <string.h>
#include <errno.h>
#include <sys/types.h>
#include <sys/inotify.h>
#include <limits.h>

#define MAX_EVENTS 1024 /*Max. number of events to process at one go*/
#define LEN_NAME 16 /*Assuming that the length of the filename won't exceed 16 bytes*/
#define EVENT_SIZE ( sizeof (struct inotify_event) ) /*size of one event*/
#define BUF_LEN ( MAX_EVENTS * ( EVENT_SIZE + LEN_NAME ) )
/*buffer to store the data of events*/

int main( int argc, char **argv )
{
    int length, i = 0, wd;
    int fd;
    char buffer[BUF_LEN], cur_dir[BUF_LEN];

    char command[100], copy_cmd[100];
    strcpy(command, "gcc -o o ");

    /* Initialize Inotify*/
    fd = inotify_init();
    if ( fd < 0 ) {
        perror( "Couldn't initialize inotify");
    }

    /* add watch to starting directory */
    wd = inotify_add_watch(fd, argv[1], IN_CREATE | IN_MODIFY);

    if ( wd == -1)
    {
        printf("Couldn't add watch to %s\n", argv[1]);
    }
    else
    {
        printf("Watching:: %s\n", argv[1]);
    }

    /* do it forever*/
    while(1)
    {
        i = 0;
        length = read( fd, buffer, BUF_LEN );

        if ( length < 0 ) {
            perror( "read" );
        }

        while ( i < length ) {
            struct inotify_event *event = ( struct inotify_event * )
```

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&buffer[ i ];
            if ( event->len ) {
                if ( event->mask & IN_CREATE ) {
                    if (event->mask & IN_ISDIR)
                        printf( "The directory %s was Created.\n", event->name );
                }
                else
                {
                    printf( "Compiling:: %s \n", event->name);

                    /*save the current dir*/
                    getcwd(cur_dir, BUF_LEN);
                    chdir(argv[1]);

                    /* form string for gcc*/
                    strcat(command, "bin/");
                    strcat(command, event->name);
                    strcat(command, ".out ");
                    strcat(command, event->name);

                    /* execute gcc*/
                    system(command);

                    /*change back to dir*/
                    chdir(cur_dir);
                    strcpy(command, "gcc -o o ");
                }
            }

            if ( event->mask & IN_MODIFY ) {
                if (event->mask & IN_ISDIR)
                    printf( "The directory %s was modified.\n", event->name );
            }
            else
            {
                printf( "Compiling:: %s \n", event->name);

                /*save the current dir*/
                getcwd(cur_dir, BUF_LEN);
                chdir(argv[1]);

                /* form string for gcc*/
                strcat(command, "bin/");
                strcat(command, event->name);
                strcat(command, ".out ");
                strcat(command, event->name);

                /* execute gcc*/
                system(command);

                /*change back to dir*/
                chdir(cur_dir);
                strcpy(command, "gcc -o o ");
            }
        }
    }
}
```